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A.C. Demagnetization of Unstable Triassic Keuper Marls from S.W. England

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Summary

In 1953, 329 specimen disks taken from 35 rock samples covering a stratigraphical thickness of 170 ft of Keuper Marl (Triassic) were collected from Sidmouth, Devonshire, and the results were described by Creer (1955, 1957a) in which an analysis of the resultant directions of natural remanent magnetization indicated the presence of three separate components, there called "*primary*", "*secondary*" and "*temporary*". Re-measurement of 65 of these rock disks from 26 of the 35 rock samples after storage in field free space for 3 years indicates that the *temporary* component of magnetization has decayed away. In further experiments, the same 65 rock disks were demagnetized in alternating magnetic fields of peak value up to 1 200 oersteds. The directions of magnetization which were originally strung out along the great circle between the primary and secondary directions progressively coalesce into two opposed groups of magnetization as the intensity of the demagnetizing field is increased. It appears that in many cases the secondary magnetization has been almost completely destroyed and in the remainder of cases considerably reduced in comparison with the primary. The method of a.c. demagnetization is described.

1. Introduction

In 1953, while at the Department of Geodesy and Geophysics, Cambridge University, the author made a detailed study of the natural remanent magnetization of samples of Keuper Marl taken over a stratigraphical thickness of 170 ft from the cliff to the east of Sidmouth, Devonshire, in S.W. England. 329 specimen disks cut from 35 samples were measured (Creer 1955, 1957a). The mean directions of magnetization of these 35 samples lay nearly along a great circle through the directions of the present axial dipole field and the mean Triassic field. The latter was calculated from the results of the work of Clegg &

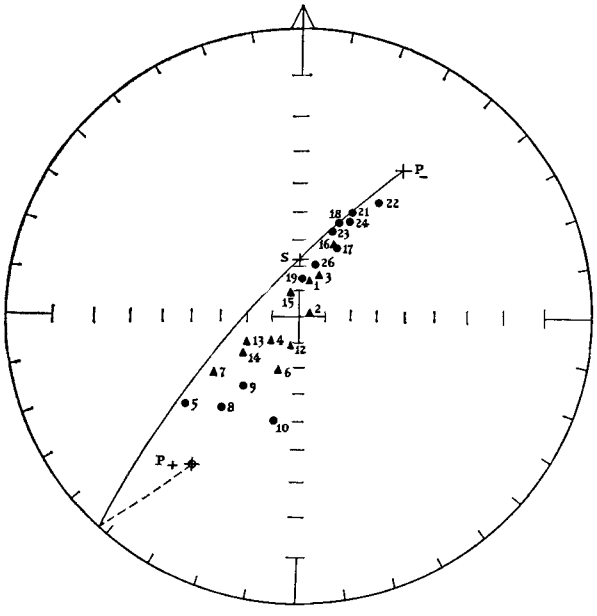


FIG. 1.—Original directions of magnetization of samples of Keuper Marl. Measurements made at Cambridge in 1953. Polar stereographic plot. Dots denote directions of north-seeking poles of n.r.m. of samples belonging to groups *N* and *R*, triangles those of samples belonging to groups *n* and *r* (see text). Sample numbers shown. Great circle P– SP_+ connects primary and secondary directions of magnetization.

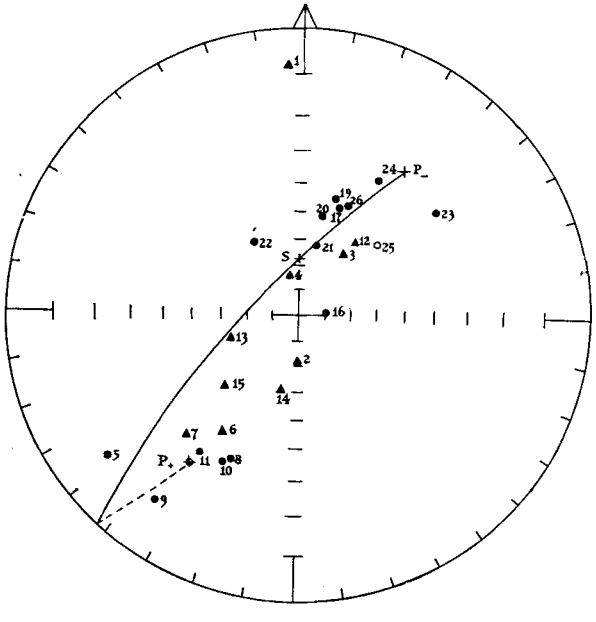


FIG. 2.—Directions of magnetization of samples of Keuper Marl made in 1958 after storage in field free space for 2 years prior to measurement. Polar stereographic plot. Convention of plotting as for Figure 1.

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others (1954) on Keuper sandstones and marls from 9 other sites in England where the natural remanent magnetization appeared stable.

The Keuper Marl would have been better named Keuper Mudstone for it is only slightly calcareous consisting of a fine silt with an admixture of clay minerals and small dolomite crystals. In many places it is structureless, closely resembling loess but at Sidmouth it is stratified at some levels probably due to deposition in temporary lakes.

When samples of this deposit are broken up a black ferromagnetic content can be separated. X-ray analysis shows this to be mostly haematite. The red cement may similarly be shown to be haematite.

2. Measurement after storage in field free space

The original measurements of directions of natural remanent magnetization made at Cambridge in 1953 are shown in Figure 1, the great circle being denoted by P_+ SP_-. For each sample the individual disk directions of magnetization have been combined in order to eliminate experimental errors and the points in Figure 1 represent mean directions for each of the 26 samples. (The term "sample" is used for each piece of rock taken from a field exposure. From each sample a number of cylindrical disks of diameter 3·75 cm and thickness 0·8 cm are cut; and measurements are made and experiments carried out on these.)

However, the inclinations of the natural remanent magnetizations are rather too steep; the points lie closer to the centre of the stereogram than the great circle. The probable explanation of this is that the disks were cut in all cases with their axes perpendicular to the bedding and were stored in the Earth's magnetic field before measurement with their bottom faces downwards but in random azimuths. Thus the vertical component of temporary magnetization was not meaned out in the original measurements, although the horizontal components were. This downward vertical component explained the effect observed in the stereogram in Figure 1.

The results obtained on remeasurement of these disks at Newcastle in 1957 after storage in zero field for about 3 years show that this explanation is satisfactory. As seen in Figure 2 the mean sample directions are now more evenly distributed on either side of the great circle P_+ SP_-.

3. Procedure of a.c. demagnetization of rocks

A.c. demagnetization has been successfully applied by Brynjolfsson (1957), Creer (1958) and As & Zijderveld (1958) to remove unstable components of magnetization of igneous rocks without appreciably affecting the primary components. To achieve this demagnetization, it has been found necessary to use a.c. fields of peak value about 100–200 oersteds which is rather less than the coercivities of the igneous rocks (see Nagata 1953). Now a back field of the same order as the coercivity is required to remove the saturated remanances and for sediments Creer (1955, 1957b) has found this to be about ten times greater varying from 2 000 to 10 000 oersteds and to be 2 700 oersteds for the Keuper Marls. Hence it was thought that demagnetizing fields up to about 2 700 oersteds would not completely destroy the primary (fossil) component, while fields of about 1 000 oersteds peak value might be necessary to demagnetize the secondary component.

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The coil system used in these experiments and in those of Creer (1958) was built and designed by Parry (1957) and consists of 2 000 turns of 17 swg wire wound in 22 layers, and has an internal diameter of 13·6 cm and an external diameter of 20·4 cm and is 13·6 cm long. It has a self inductance of 0·4 henries and is tuned to the 50 c/s mains supply by connecting in series with a bank of condensers, the exact capacitance being determined by trial and error during tuning tests. The coil was calibrated by measuring the output of a small search coil placed at its centre and was found to produce 200 oersteds (peak value) per ampere (r.m.s.). The 240 V mains will give a maximum current of 10 A (r.m.s.) and hence the coil is capable of producing an alternating magnetic field of 2 000 oersteds peak value.

Clegg (private communication) and As & Zijderveld (1958) report random magnetization of some specimens after a.c. demagnetization and claim that this is due to the presence of unsymmetrical even harmonics in the mains supply, producing a magnetization in the specimens so treated. Creer (1958) demagnetized specimens about three mutually perpendicular axes in both directions and this effect was not marked. However in strong alternating magnetic fields, results were not always entirely satisfactory so a device was built in which 4 sedimentary rock disks 3·5 cm diameter and 0·8 cm thick or 1 igneous core 3·5 cm diameter and 3·5 cm thick could be rotated simultaneously about two perpendicular axes while in the demagnetizing field. These axes are both perpendicular to the axis of the coil and have speeds of rotation in the ratio 1:1·1 so that any direction in the sample performs a Lissajous figure during rotation and all directions in the rock are demagnetized. The speed of rotation can be varied between about 1 and 3 revolutions per second, which is roughly 50 to 20 times slower than the frequency of the main current supply. This device thus prevents the specimens acquiring an i.r.m. due to d.c. components of the origin mentioned above. Care was taken to ensure that the speeds of rotation of the specimen were not sub-harmonics of 50 c/s.

The normal practice in a.c. demagnetization is slowly to decrease the demagnetizing field to zero from the chosen maximum value. It is important that the rate at which this field is decreased is even and slow compared with the speed of rotation of the specimen. To ensure a smooth and slow fall off in demagnetizing field an electrolytic resistance has been introduced in the circuit. It consists of a tube 10 cm in diameter and about 1 m long containing copper sulphate solution. One copper electrode is fixed to the bottom and another to a probe which is gradually pulled upwards through the electrolyte by means of a string and pulley system powered by an electric motor which turns at 30 revolutions per hour. It is arranged, by gradually increasing the speed of withdrawal of the probe as the electrode separation is increased, that the resistance of the electrolyte increases at a uniform rate and hence the a.c. current through the coil is reduced steadily and uniformly. It requires 10 min to reduce the demagnetizing field from about 1 000 oersteds to zero, and the value of the current passing through the demagnetizing coil is registered on a pen recorder. A photograph of the apparatus is shown in Figure 3.

In earlier experiments in which the demagnetizing field was gradually decreased to zero by slowly pulling the coil away from the specimen along non-magnetic roller bearings, it proved impossible to withdraw the coil sufficiently slowly and smoothly because of uneven friction. (The field intensity drops by a factor of about $\frac{5}{8}$ during the initial movement of 6·8 cm, the half length of the coil.) Hence this procedure was abandoned in favour of the variable electrolytic resistance, but the

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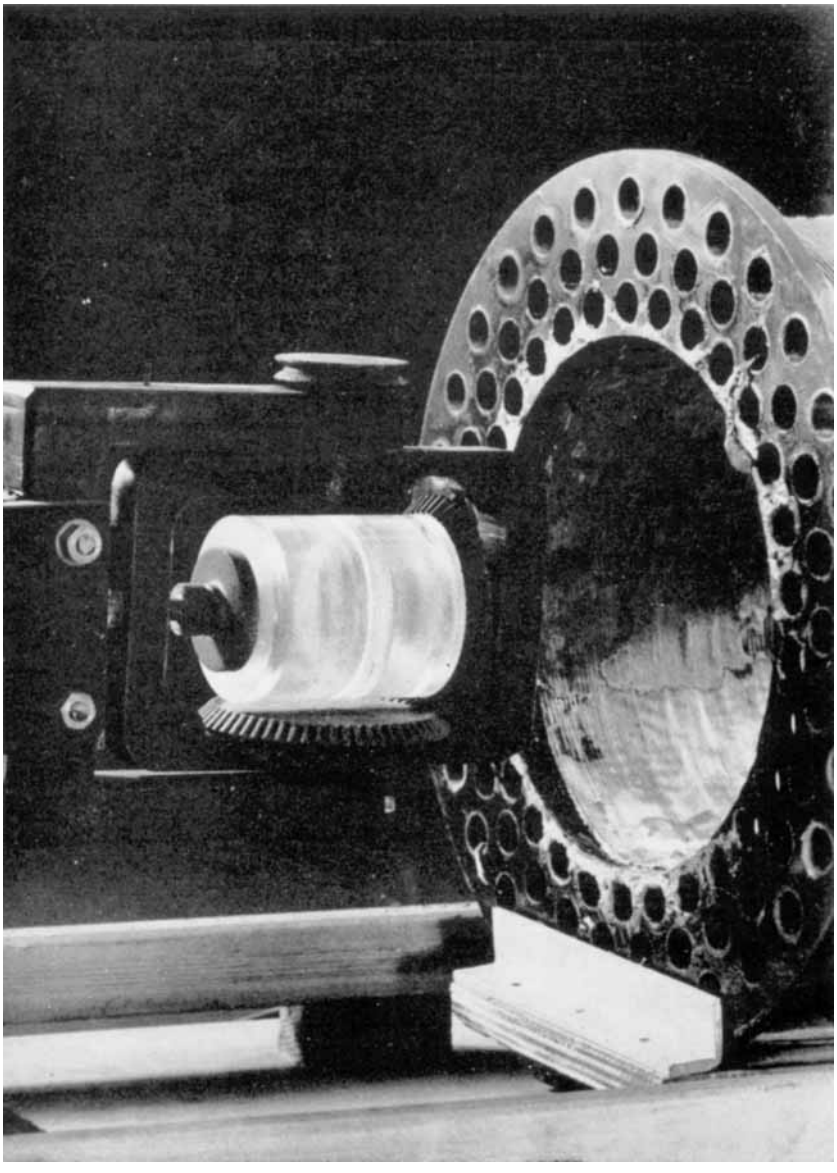


FIG. 3.—Photograph of demagnetizing coil and specimen holder.

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latter is not satisfactory when used for more than a few minutes at a time when passing currents of greater than 6 A and this limited the intensity of a.c. field applied in these experiments to 1 200 oersteds. It has, however, since been found possible to reduce the field sufficiently smoothly by suspending a cradle 1 m long containing the coil from the ceiling by six threads such that it can move only in a direction parallel to its axis and slowly pulling it away automatically.

4. Demagnetization experiments on Keuper Marls

4.1. Experimental procedure

The 65 rock disks were demagnetized by alternating magnetic fields of peak value 400, 600, 800, 1 000 and finally 1 200 oersteds. The mean direction of magnetization of each of the 26 rock samples was found by averaging the directions of magnetization of its 2 or 3 component disks, the effect of stray magnetizations acquired during "cleaning" being reduced considerably in this way. Stereographic projections of the results are shown in Figures 4 to 9.

4.2. Classification of results

An inspection of Figure 2 and Figures 4 to 9 shows that it is possible to classify the rock samples into four groups: (i) *group N* consisting of samples the directions of magnetization of which are centred about the mean direction of magnetization P_- of stable normally magnetized Triassic rocks, (ii) *group R* consisting of samples the directions of magnetization of which progressively coalesce around the mean direction of magnetization P_+ of stable reversely magnetized Triassic rocks as the value of the demagnetizing field is increased and (iii) *groups n and r* consisting of samples which are apparently not completely cleaned by the treatment described. Included here also are two samples the directions of magnetization of which do not change in an obviously systematic manner.

In Figures 2, 4, 5, 6, 7, 8 and 9 points representing directions of magnetization of groups *N* and *R* are shown by dots or open circles and those of groups *n* and *r* by full or open triangles.

Paths traced by successive directions of magnetization of samples at 0, 400, 600, 800, 1 000 and 1 200 oersteds respectively during the experiments are shown in Figure 13. Theoretically, if the directions are resultants of primary and secondary components these paths should follow great circles. That they do not is due to stray magnetizations picked up during the cleaning process rather than to errors in measurement. One marked example of this is given by sample 12 (group *r*) at 1 000 oersteds. Sample 26 (group *N*) shows no significant change in direction, though group *N* as a whole shows a significant change in average direction of magnetization during the experiment as is shown in Figure 11. Sample 2 (possibly group *n*) shows no significant change, though sample 10 (group *R*) gives a path approximating to the great circle P_-SP_+.

4.3. Discussion of results

4.3.1. *Stratigraphical succession of samples.*—The rock samples have been numbered in order of stratigraphical succession as indicated in Table 1, from which the measurements described in this paper may be correlated with the original measurements published by Creer (1957a).

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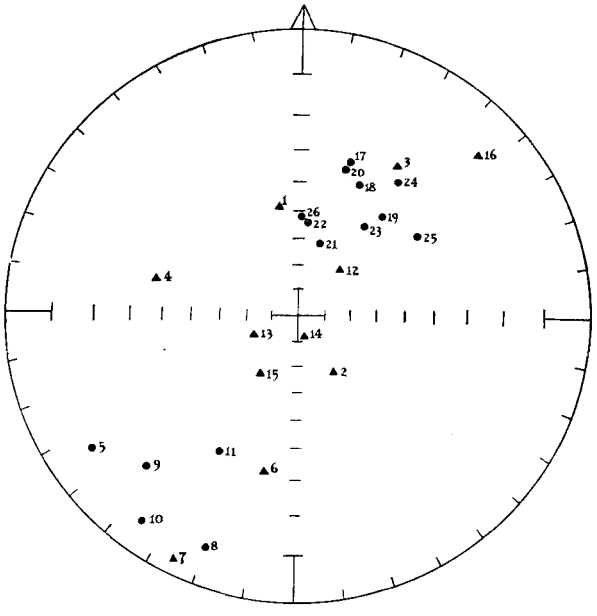


FIG. 4.—Directions of magnetization of samples of Keuper Marl after demagnetization in a.c. magnetic field of peak value 400 oersteds. Dots and full triangles represent directions of north-seeking poles of remanent magnetization with downward inclinations, circles and hollow triangles, directions of north-seeking poles of remanent magnetization with upward inclinations.

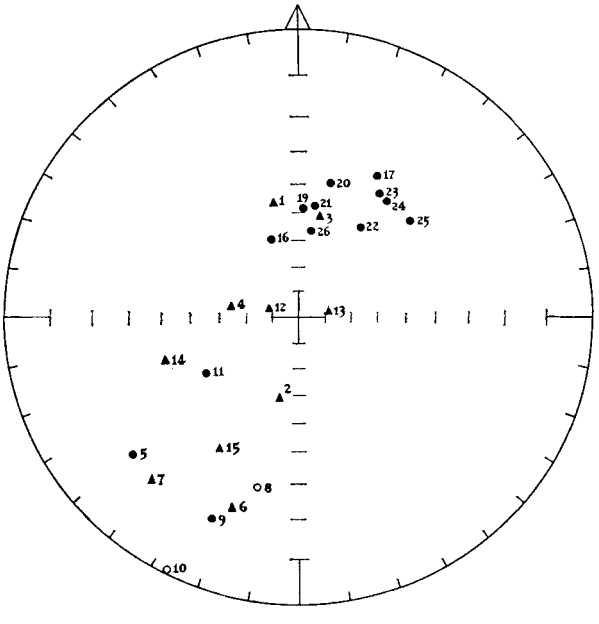


FIG. 5.—Directions of magnetization of Keuper Marl samples after demagnetization in a.c. magnetic fields of peak value 600 oersteds. Plotting convention as for Figure 4.

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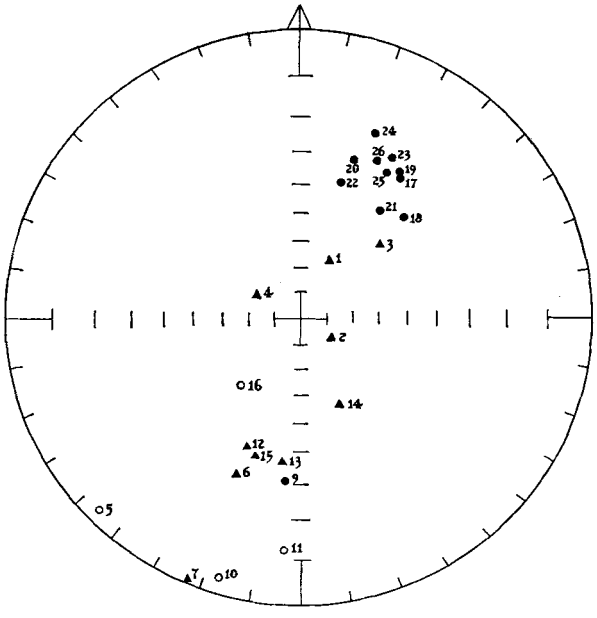


FIG. 6.—Directions of magnetization of Keuper Marl samples after demagnetization in a.c. magnetic fields of peak value 800 oersteds. Plotting convention as for Figure 4.

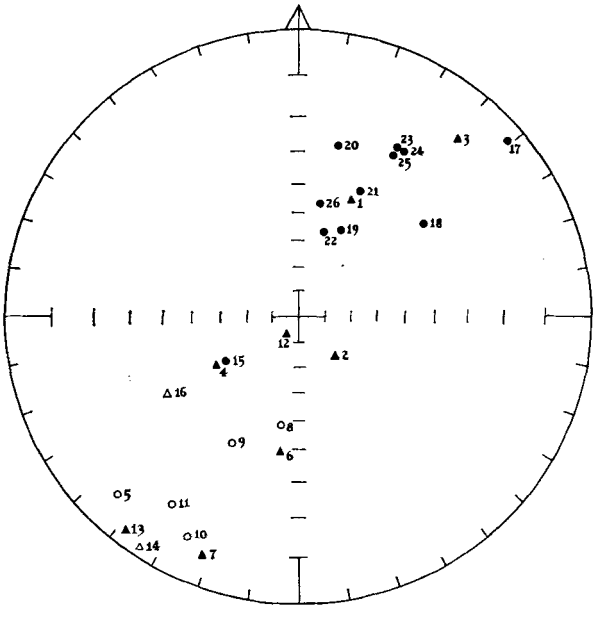


FIG. 7.—Directions of magnetization of Keuper Marl samples after demagnetization in a.c. magnetic fields of peak value 1 000 oersteds. Plotting convention as for Figure 4.

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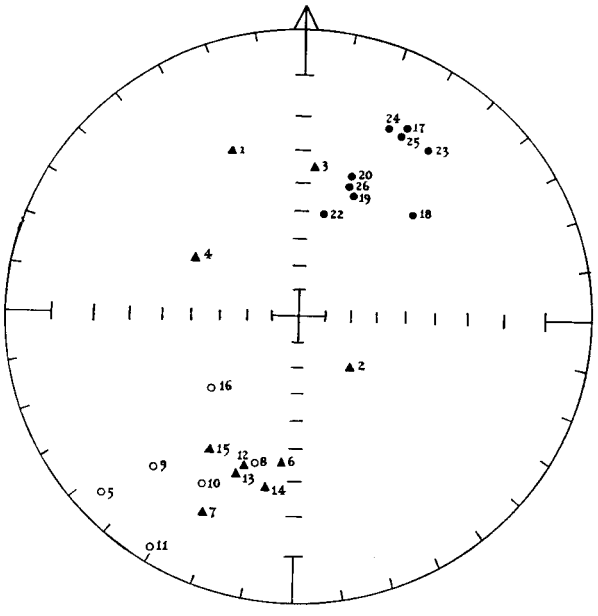


FIG. 8.—Directions of magnetization of Keuper Marl samples after demagnetization in a.c. magnetic fields of peak value 1 200 oersteds. Plotting convention as for Figure 4.

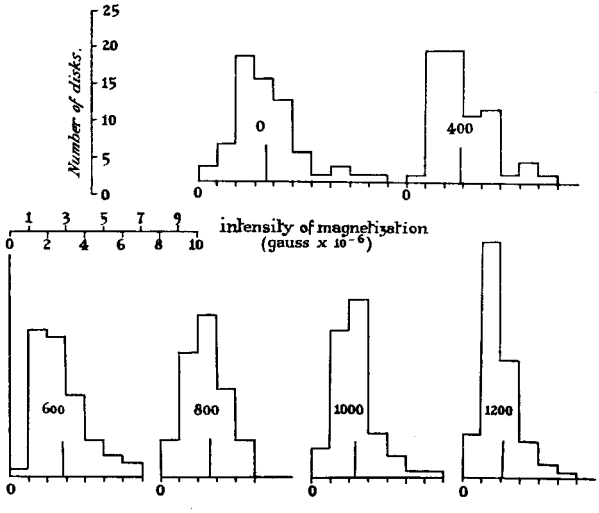


FIG. 9.—Histograms of intensities of magnetization of rock disks from samples of Keuper Marl at successive stages of the experiment. Mean intensity shown by vertical line drawn inside each histogram.

Those samples falling into groups *R* and *r* are confined to the middle 36·5 ft of the cliff, those falling in group *N* to the upper 36 ft. This group may continue stratigraphically upwards beyond the topmost horizon sampled because a sample from the 170 ft horizon previously studied (Creer 1957a) had a virgin direction of magnetization N 28 E + 40°. Samples falling into group *n* come from the bottom 35 ft of the cliff.

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Table 1

Stratigraphic level	Sample number	Direction of magnetization after cleaning
145 ft	26	*N*
143·5	25	*N*
143	24	*N*
142·5	23	*N*
219	22	*N*
127·5	21	*N*
115·5	20	*N*
111·5	19	*N*
111·0	18	*N*
109·5	17	*N*
108·5	16	*R*
106	15	*r*
105	14	*r*
103	13	*r*
101	12	*r*
91·5	11	*R*
91·0	10	*R*
90·5	9	*R*
90	8	*R*
89·5	7	*r*
88·5	6	*r*
77	5	*R*
72	4	?
35	3	*n*
15	2	?
0	1	*n*

Samples comprising groups N and R come from the well stratified beds which appear to be water-deposited in contrast to those from groups n and r which come from the more massive loess type beds.

4.3.2. *Intensity histograms.**—Figure 9 consists of histograms of the intensities of individual disks at each of the stages of the experiment. The mean intensity of magnetization for all the samples is also shown for each stage and it may be observed first that it decreases monotonically as the strength of the demagnetizing field is increased, and second that the deviation about the mean direction also decreases as the field is increased.

4.3.3. *Groups N and R.*—Mean intensities of magnetization at successive stages of the experiment have been calculated for samples 17–26 inclusive comprising group N and for samples 5 and 8–11 from group R. These, with their standard deviations σ and standard deviations of the mean (standard errors) σ_m are listed in Table 2. The mean intensities of magnetization for each of groups N and R are plotted as functions of the peak value of demagnetizing field in Figure 10 from which it is seen that the intensities of magnetization of groups N and R are not significantly different from one another after cleaning although the intensity of magnetization of group N was considerably greater than that of group R, after

* Due to an error, all intensities of magnetization given in Creer (1957a) for the original magnetization are too low by a factor of 10.

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Table 2

Mean intensities of magnetization of Keuper Marl samples on demagnetization

units: gauss $\times 10^{-6}$

Peak value of demagnetizing field (oersteds)	Normal (Group N)			Reversed (Group R)			Others (Groups n and r)		
	M	σ	σ_m	M	σ	σ_m	M	σ	σ_m
0	4·37	1·15	0·24	2·98	1·11	0·37	3·45	2·38	0·47
400	3·77	1·33	0·27	2·37	0·75	0·23	2·51	1·35	0·27
600	3·02	1·23	0·26	2·63	0·76	0·23	2·65	1·46	0·29
800	2·89	0·92	0·19	2·43	0·54	0·16	1·82	0·94	0·20
1 000	2·54	1·18	0·25	2·39	0·77	0·22	2·04	1·18	0·26
1 200	2·22	1·00	0·21	2·39	0·88	0·25	1·77	0·69	0·14

storage in field free space and before cleaning. Mean directions of magnetization of samples from groups N and R have been calculated after each demagnetization. These are shown in Figure 11 where S, the assumed direction of the secondary

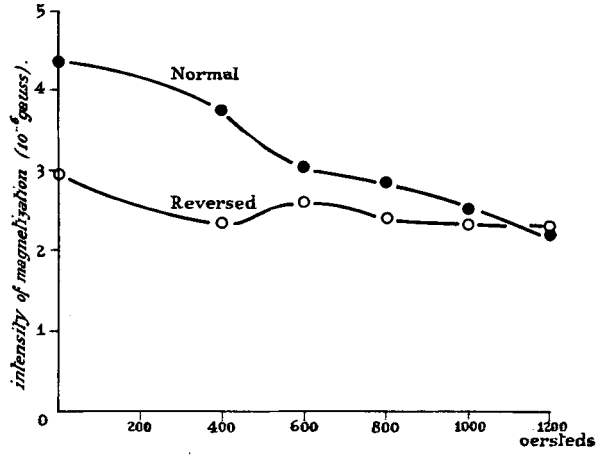


FIG. 10.—Average intensities of magnetization of all normally magnetized samples (17 to 26, i.e. group N) and of reversely magnetized samples (5, 8, 9, 10 and 11, from group R) at successive stages of the experiment.

component, is that of a geocentric axial dipole field at Sidmouth and P_- and P_+ the assumed normal and reversed directions of the primary component are the poles of the mean axis of magnetization of stable Keuper sandstones studied by Clegg & others (1954). Successive mean directions of magnetization of both groups move along paths close to the great circle P_- SP_+ indicating that the secondary component is demagnetized more readily than the primary. The angles between the resultant directions of magnetization and the assumed primary and secondary directions were measured graphically and the intensities of the

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primary and secondary components were then calculated for each stage. In Figure 12 these are shown as functions of the peak value of the demagnetizing field. From these graphs the secondary component is seen to be much "softer" than the primary, the former being almost completely destroyed by a.c. fields of 800–1 000 oersteds and the latter retaining about three-quarters of its initial value after having been subjected to demagnetization in an a.c. field of 1 200 oersteds. Assuming that the angles between the resultant directions of magnetization and P_ and S respectively can be estimated graphically to 2°, the estimated error for the relative intensities of components *P* and *S* works out at about 15 per cent. This probably explains the scatter of the *P* and *S* intensities shown in Figures 12a and 12b.

The successive changes in direction of magnetization of some individual samples from groups *N* and *R* are shown in Figure 13.

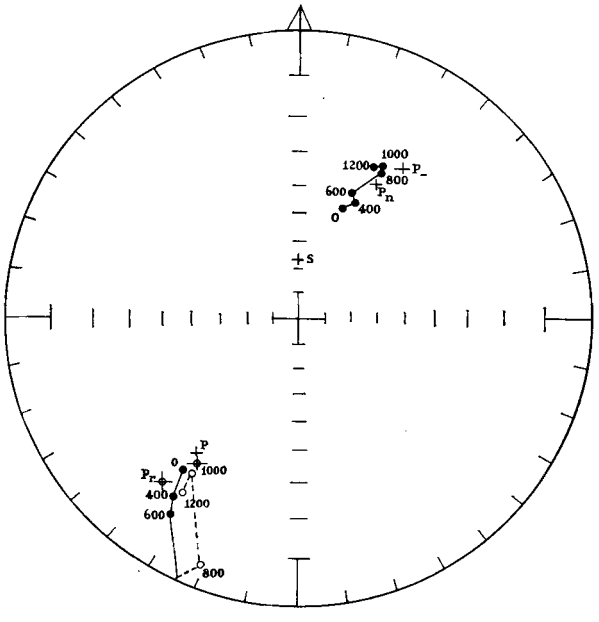


FIG. 11.—Paths followed by average directions of magnetization of groups *N* and *R* during the experiment. P_-, P_+ normal and reversed directions of Triassic axis of magnetization. P_n and P_r mean directions of magnetization of normally and reversely magnetized Keuper Sandstones studied by Clegg & others (1954). S direction of dipole field along present geographic axis at Sidmouth.

4.3.4. *Group r.*—The samples of this group come from the middle of the cliff. The directions of magnetization after storage in zero field (Figure 2) are all steeply inclined to the bedding planes, close to the great circle P_-SP_+ and near S, the direction of the secondary component. This suggests that the ratios between the intensities of the secondary and primary components are greater than in groups *N* and *R*. As the cleaning became progressively more severe the directions of magnetization of these samples moved along paths close to the great circle P_-SP_+ towards P_+ (for examples see Figure 13), but showed no sign of reaching the point P_+. Further cleaning in demagnetizing fields higher than 1 200 oersteds might align the resultant directions of magnetization of these samples more closely parallel to P_+ as in group *R*. However, the intensity of magnetization is generally

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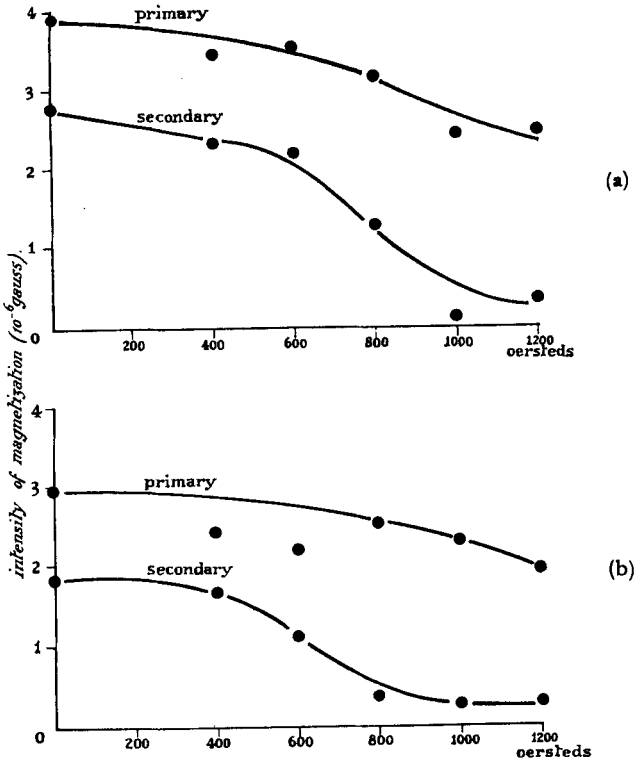


FIG. 12a.—Demagnetization curves for primary and secondary components of group *R* (reversed).

FIG. 12b.—Demagnetization curves for primary and secondary components of group *N* (normal).

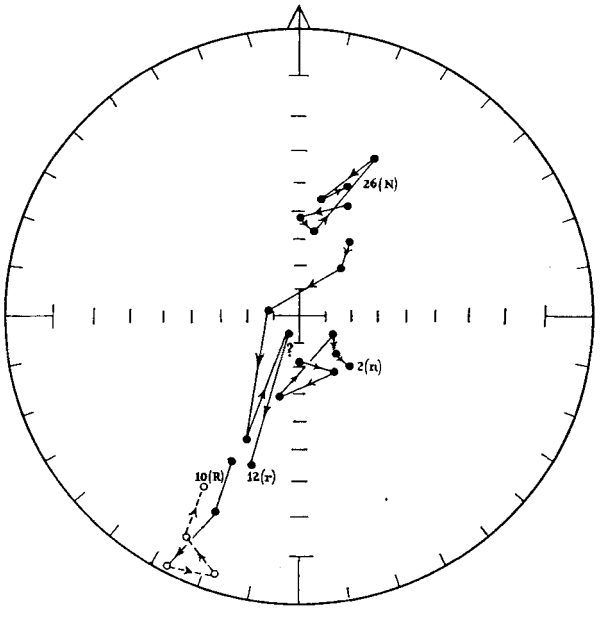


FIG. 13.—Successive directions of magnetization of representative samples (numbers and groups shown) after treatment in peak demagnetizing fields of 0, 400, 600, 800, 1 000 and 1 200 oersteds respectively.

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lower than in group *R* and repeated readings of intensity and direction for disks subjected to the same treatment more than once are not so good as for group *R*. It is possible that the directions of magnetization after cleaning in fields of 1 000 or 1 200 oersteds represent the direction of the geomagnetic field when these beds become magnetized, but this is improbable because of the poor repeatability and because the *rR* zone is bounded at its upper and lower limits by *R* type samples (see Table 1). That samples of group *r* come from the loess type beds is the most probable reason for their different experimental behaviour from samples from group *R* which come from thinner water laid beds.

4.3.5. *Group n.*—Samples 1 and 3 show similar characteristics to those of group *N*, their directions of magnetization tending to move towards P_ as the demagnetizing field is increased. However, repeated measurements following further treatment in the same demagnetizing field lack repeatability as in group *r*. Samples 2 and 4 show no significant change in direction of magnetization on cleaning, although both suffered a drop in intensity to about half the initial value. From their response to experimental treatment alone it is difficult to decide whether these two samples should be placed in group *n* or *r*, though from their stratigraphical position group *n* is more likely.

Samples 1, 2, 3 and 4 all come from loess type beds.

Table 3

Directions of magnetization, standard deviations θ_0 and standard errors θ_m

Demagnetizing field (peak value in oersteds)	Normal (Group *N*)				Reversed (Group *R*)			
	D	*I*	θ_0	θ_m	*D*	*I*	θ_0	θ_m
0	N 22° E	+46°	14°·7	5°·2	S 36° W	+22°	15°·3	8°·8
400	N 26° E	+43°	16°·8	5°·6	S 34° W	+15°	19°·6	9°·8
600	N 25° E	+40°	14°·5	5°·1	S 32° W	+11°	32°·8	16°·4
800	N 29° E	+30°	9°·2	3°·1	S 18° W	+ 3°	26°·2	15°·2
1 000	N 29° E	+29°	19°·9	6°·6	S 34° W	−23°	20°·7	10°·3
1 200	N 27° E	+30°	14°·8	5°·2	S 33° W	−17°	16°·9	8°·4
Mean value for stable Triassic rocks	N 29° E	+34°	—	—	S 39° W	−16°	—	—

5. Conclusions

Demagnetization experiments on Keuper Marl samples have confirmed that the natural magnetization of these rocks consists of a *primary* component in the same direction as the stable n.r.m. of normally or reversely magnetized Keuper Sandstone together with a magnetically softer *secondary* component. The latter requires an alternating field of peak value greater than 800 oersteds to destroy it, and is therefore considerably harder than the soft component of unstable igneous rocks which is completely demagnetized by a.c. fields of peak value 100–200

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oersteds. It is also considerably harder than the stable magnetization found in igneous rocks which is reduced to roughly half by a.c. fields of peak value 300–400 oersteds (Creer 1958).

The primary component of this sediment is extremely hard and retains about three-quarters of its natural value after demagnetization in a.c. fields of peak value 1 200 oersteds.

The 145 ft of Keuper Marl studied appear to consist of an upper normally magnetized zone more than 35 ft and possibly more than 60 ft thick, a central reversely magnetized zone of thickness 36·5 ft and possibly a lower normally magnetized zone. No chemical or mineralogical differences between the normally and reversely magnetized samples have been detected, both types of zone containing water-laid and loess type deposits.

If the reversed magnetization is due to a reversal of the main geomagnetic field it should be possible to infer something of the way the reversal takes place and of the time required for a reversal to occur. However, although the estimated stratigraphical separation between the top sample (No. 16) of the reversed zone and the bottom sample (No. 17) of the upper normal zone is only 1 ft these samples come from different beds. Further, a horizontal distance of 10 ft separated samples 16 and 17 and this was covered by scree. A fault between these two points appeared unlikely but the possibility of a small fault of a few feet throw must be borne in mind and therefore the time taken for the reversal to occur cannot be estimated. Also there is no way of estimating the time interval between the deposition of the beds.

The transition between the lower normal zone and the base of the reversed zone does not appear to be nearly so sharp, though sampling at this horizon was not carried out so intensively and the loess type of rock encountered here is not very suitable for palaeomagnetic work because of the possible existence of a comparatively hard secondary component.

The thickness of a complete reversed zone of about 36 ft is probably accurate only to 5 ft as the beds could not always be followed along the base of the cliff due to rock falls. However, no fault of more than a few feet was encountered. Little is known of deposition rates of sediments and no geologist acquainted with this formation is prepared to venture a reliable estimate of the time required for the deposition of this 35 ft. Further, since the origin of magnetization was probably chemical, rates of deposition may not have any direct bearing on the time required for magnetization to take place.

Further field work is planned in this region to study in more detail the horizons in which the reversals occur. When the field work was originally done in 1953 the intention was merely to measure the mean direction of n.r.m. of this formation and for the more detailed studies now planned it will be necessary to make much more thorough collections. More detailed studies on the exposures of stable Keuper sandstones visited by Clegg & others (1954) are also planned by the present author as in their original work only one or two rock samples were collected from each locality. A more intense sampling over a considerable stratigraphical thickness might possibly reveal a similar pattern of reversals to those found at Sidmouth.

If this occurs and if the reversals are not of equal duration it may be possible to correlate the different isolated outcrops of Keuper rocks in England, which is not possible by normal geological methods.

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